BIG DATA (Compiled by AMRITA)

UNIT 1

Introduction to Big data

Big data refers to massive, complex datasets that traditional data management systems cannot handle. When properly collected, managed and analyzed, big data can help organizations discover new insights and make better business decisions. While enterprise organizations have long collected data, the arrival of the internet and other connected technologies significantly increased the volume and variety of data available, birthing the concept of “big data”. Today, businesses collect large amounts of data often measured in terabytes or petabytes on everything from customer transactions and social media impressions to internal processes and proprietary research. Data science and more specifically, big data analytics help organizations make sense of big data’s large and diverse datasets. These fields use advanced tools such as machine learning to uncover patterns, extract insights and predict outcomes.
In recent years, the rise of artificial intelligence (AI) and machine learning has further increased the focus on big data. These systems rely on large, high-quality datasets to train models and improve predictive algorithms.

Traditional data and big data differ mainly in the types of data involved, the amount of data handled and the tools required to analyze them.
Traditional data primarily consists of structured data stored in relational databases. These databases organize data into clearly defined tables, making it easy to query through standard tools like SQL. Traditional data analytics typically involves statistical methods and is well suited for datasets with predictable formats and relatively small sizes.

Big data, by contrast, encompasses massive datasets in various formats, including structured, semi-structured and unstructured data. This complexity demands advanced analytical approaches—such as machine learning, data mining and data visualization—to extract meaningful insights. The sheer volume of big data also requires distributed processing systems to handle the data efficiently at scale.

Types of Big Data

Structured data

Definition: Information organized into a rigid, predefined schema of rows and columns, similar to a spreadsheet.

Characteristics: Quantitative, highly searchable, numeric/categorical, and easy to map.

Examples: SQL relational databases, customer relationship management (CRM) records, financial transactions, inventory logs, dates, and phone numbers.

Storage & Tools: Relational databases and data warehouses like MySQL, PostgreSQL, Oracle, Snowflake, or Google BigQuery; queried using

Unstructured Data

Definition: Information that has no predefined data model, schema, or tabular organization.

Characteristics: Qualitative, complex, makes up 80% to 90% of all enterprise data, and difficult to search with traditional methods.

Examples: Word documents, PDFs, audio/video recordings, digital photos, social media posts, satellite imagery, and call center transcripts.

Storage & Tools: Data lakes and object storage platforms (like Amazon S3 or Google Cloud Storage); analyzed using Natural Language Processing (NLP), computer vision, and Generative AI/RAG tools

Semi-Structured Data

Definition: A bridge between structured and unstructured data; it lacks a fixed relational table schema but uses tags, markers, or key-value pairs to enforce internal hierarchies.

Characteristics: Flexible, self-describing, and contains a mix of categorical tags and free-form content.

Examples: JSON files, XML, YAML, HTML, server/IoT logs, and emails (which contain structured headers and unstructured body text).

Storage & Tools: NoSQL databases (like MongoDB, Apache Cassandra, or Google Firestore) and data lakes using schema-on-read

The V's of big data

The "V's of Big Data"—volume, velocity, variety, veracity and value—are the five characteristics that make big data unique from other kinds of data. These attributes explain how big data differs from traditional datasets and what's needed to manage it effectively.

Volume

Big data is "big" because there's more of it. The massive amount of data generated today—from web apps, Internet of Things (IoT) devices, transaction records and more—can be hard for any organization to manage. Traditional data storage and processing systems often struggle to handle it at scale. Big data solutions, including cloud-based storage, can help organizations store and manage these ever-larger datasets and make sure that valuable information is not lost to storage limits.

Velocity

Velocity describes how quickly data flows into a system. Big data emphasizes the rapid pace at which this data moves.Today, data arrives faster than ever, from real-time social media

updates to high-frequency stock trading records. This rapid data influx provides opportunities for timely insights that support quick decision-making. To handle this trend, organizations use tools like stream processing frameworks and in-memory systems to capture, analyze and act on data in near real-time.

Variety

Variety refers to the many different formats that big data can take.Along with traditional structured data, big data can include unstructured data, such as free-form text, images and videos. It can also include semi-structured data, such as JSON and XML files, that have some organizational properties but no strict schema.Managing this variety requires flexible solutions like NoSQL databases and data lakes with schema-on-read frameworks, which can store and integrate multiple data formats for more comprehensive data analysis.

Veracity

Veracity refers to the accuracy and reliability of data. Because big data comes in such great quantities and from various sources, it can contain noise or errors, which can lead to poor decision-making.Big data requires organizations to implement processes for ensuring data quality and accuracy. Organizations often use data cleaning, validation and verification tools to filter out inaccuracies and improve the quality of their analysis.

Value

Value refers to the real-world benefits that organizations gain from big data. These benefits include everything from optimizing business operations to identifying new marketing opportunities. Big data analytics is critical for this process, often relying on advanced analytics, machine learning and AI to transform raw information into actionable insights.

The evolution of big data

The concept of big data first emerged in the mid-1990s when advances in digital technologies meant organizations began producing data at accelerated rates. Initially, these datasets were smaller, typically structured and stored in traditional formats.However, as the internet grew and digital connectivity spread, big data was truly born. An explosion of new data sources, from online transactions and social media interactions to mobile phones and IoT devices, created a rapidly growing pool of information.This surge in the variety and volume of data drove organizations to find new ways to process and manage data efficiently.

Benefits of big data

Big data has transformed how organizations gather insights and make strategic decisions. A study by Harvard Business Review found that data-driven companies are more profitable and innovative than their peers.Organizations effectively employing big data and AI reported outperforming their peers in key business metrics, including operational efficiency (81%

versus 58%), revenue growth (77% versus 61%) and customer experience (77% versus 45%).

Here are some of big data's most significant benefits and use cases.

Improved decision-making: Analyzing vast datasets allows organizations to uncover patterns and trends that lead to more informed decisions. For instance, a grocery chain can use sales data and weather forecasts to predict demand for seasonal products, helping to stock stores and reduce waste.

Enhanced customer experience: Big data enables companies to understand customer behavior at a more granular level, paving the way for more tailored interactions. For instance, big data analytics can help identify customers who frequently buy skincare products from a specific brand. The brand can use this information to help target campaigns for limited-time sales or special offers on similar products.

Increased operational efficiency: Real-time data allows organizations to streamline operations and reduce waste. In manufacturing, for example, organizations can analyze real-time sensor data to predict equipment failures before they occur. This process, known as predictive maintenance, can help prevent downtime and reduce maintenance costs.

Responsive product development: Big data insights help companies respond to customer needs and guide product improvements. For example, if multiple users report that a specific feature in a smartphone drains battery life too quickly, developers can prioritize optimizing that feature in the next software update.

Optimized pricing: Big data enables organizations to refine pricing strategies based on real-time market conditions. For example, an airline can use insights derived from big data to adjust ticket prices dynamically, responding to demand shifts and competitor pricing.

Enhanced risk management and fraud detection: Big data allows organizations to identify and monitor risks proactively. Banks, for instance, analyze transaction patterns to detect potential fraud. If a customer's credit card is used for an unusual high-value purchase in another country, the bank can flag the transaction and notify the customer for verification.

Healthcare innovation: Healthcare providers can use big data to make sense of patient records, genetic information and data from wearable devices. For example, a continuous glucose monitor for a patient with diabetes can track blood sugar levels in real time. This capability allows healthcare providers to detect dangerous spikes or drops and adjust treatment plans as needed.

Challenges of big data

While big data offers immense potential, it also comes with significant challenges, especially around its scale and speed.

Some of the biggest challenges of big data include:

Data quality and management: Connecting datapoints and keeping data accurate can be a complex undertaking especially with massive amounts of information constantly streaming in from social media, IoT devices and other sources. For example, a logistics company might struggle to integrate GPS data from its fleet with customer feedback and warehouse inventory to get a precise view of delivery performance.

Scalability: As data grows, organizations must expand storage and processing systems to keep up. For instance, a streaming platform analyzing millions of daily viewer interactions often needs to constantly add to its storage and compute power to handle the demand. Cloud services can offer more scalable alternatives to on-premises solutions, but managing high volumes and velocities of data can still be difficult.

Privacy and security: Regulations like GDPR and HIPAA require strict data privacy and security measures, such as strong access controls and encryption to prevent unauthorized access to patient records. Complying with these mandates can be tough when datasets are massive and constantly evolving.
Integration complexity: Combining different types of data from multiple sources can be technically demanding. For instance, a retail chain might struggle to consolidate structured sales records with unstructured customer reviews and semi-structured supplier data for a comprehensive view of product performance.
Skilled workforce: Big data work requires specialized skills in data science, engineering and analytics. Many organizations face ongoing challenges finding professionals like data analysts and other specialists who can manage and interpret large datasets. For example, a financial institution might struggle to hire data scientists skilled in both machine learning and financial modeling to analyze transaction data and predict market trends.

Unit 2

Apache Hadoop

Apache Hadoop is an open-source java based software framework developed by Douglas Cutting, then at Yahoo, that provides the highly reliable distributed processing of large data sets using simple programming models. It is designed to scale-up from single servers to thousands of machines, each providing computation and storage, this gives distributed storage and distributed processing.
The main goals of Hadoop are listed below:
1. Scalable: It can scale up from a single server to thousands of servers.
2. Fault tolerance: It is designed with very high degree of fault tolerance.
3. Economical: It uses commodity hardware instead of high-end hardware.
4. Handle hardware failures: The resiliency of these clusters comes from the software's ability to detect and handle failures at the application layer.
The Hadoop framework can store huge amounts of data by dividing the data into blocks and storing it across multiple computers, and computations can be run in parallel across multiple connected machines.

Hadoop consists of the following components:
1. Hadoop Common: This package provides file system and OS level abstractions. It contains libraries and utilities required by other Hadoop modules. This consists of necessary Java archive (JAR) files and scripts needed to start Hadoop. Hadoop requires Java Runtime Environment (JRE) 1.6 or higher version. The standard start-up and shut-down scripts need Secure Shell (Ssh) to be setup between the nodes in the cluster.

2. Hadoop Distributed File System (HDFS): {STORAGE}HDFS is a distributed file system that provides a limited interface for managing the file system.

3. Hadoop MapReduce: {PROCESSING}MapReduce is the key algorithm that the Hadoop MapReduce engine uses to distribute work around a cluster.

4. Hadoop Yet Another Resource Negotiator (YARN) (MapReduce 2.0): It is a resource-management platform responsible for managing compute resources in clusters and using them for scheduling of users' applications.

Hadoop Distributed File System (HDFS)

HDFS creates multiple replicas of each data block and distributes them on computers throughout a cluster to enable reliable and rapid access. When a file is loaded into HDFS, it is replicated and fragmented into “blocks” of data, which are stored across the cluster nodes; the cluster nodes are also called the DataNodes. The NameNode is responsible for storage and management of metadata, so that when MapReduce or another execution framework calls for the data, the NameNode informs it where the data that is needed resides.

Main Components of HDFS

NameNode is the master that contains the metadata. In general, it maintains the directories and files and manages the blocks which are present on the DataNode. The following are the functions of NameNode:

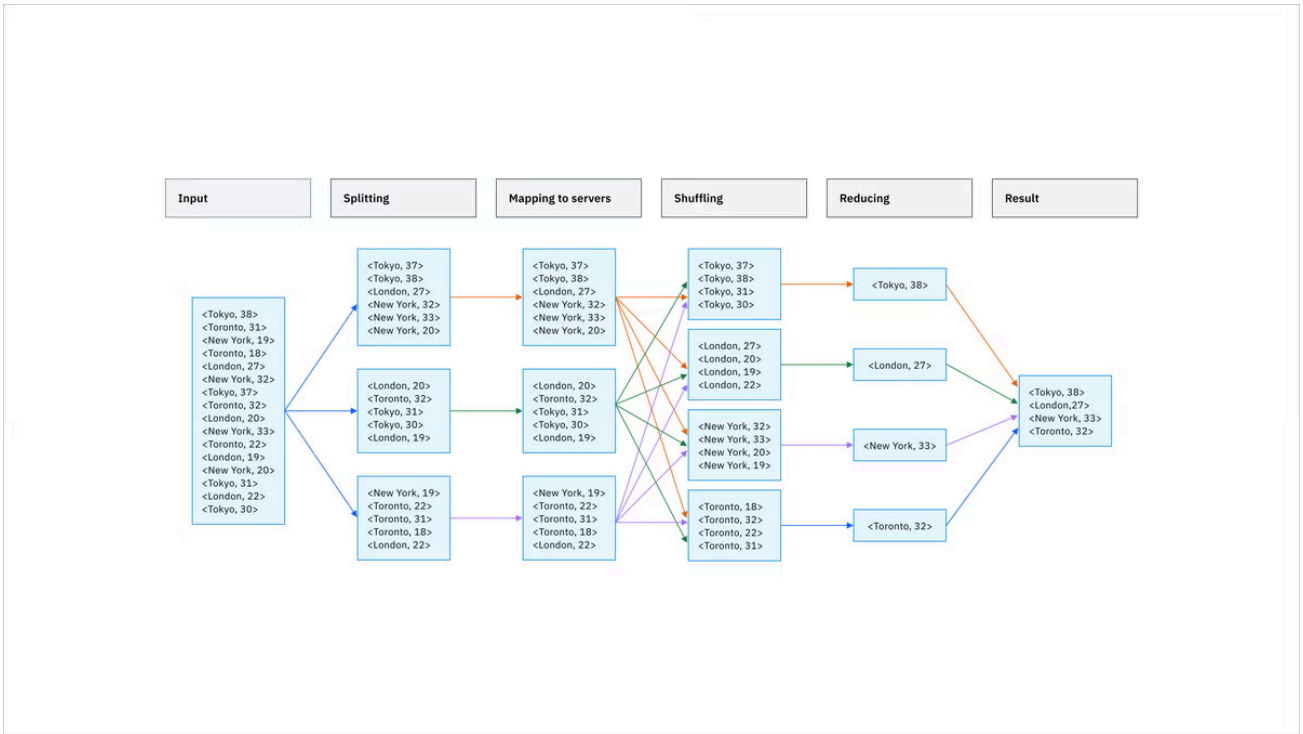
1. Manages namespace of the file system in memory.
2. Maintains “inode” information.
3. Maps inode to the list of blocks and locations.
4. Takes care of authorization and authentication.
5. Creates checkpoints and logs the namespace changes.

So the NameNode maps DataNode to the list of blocks, monitors status (health) of DataNode and replicates the missing blocks.

DataNodes are the slaves which provide the actual storage and are deployed on each machine. They are responsible for processing read and write requests for the clients. The following are the other functions of DataNode:

1. Handles block storage on multiple volumes and also maintain block integrity.
2. Periodically sends heartbeats and also the block reports to NameNode.

MapReduce The MapReduce algorithm aids in parallel processing and basically comprises two sequential phases: map and reduce.
1. In the map phase, a set of key–value pairs forms the input and over each key–value pair, the desired function is executed so as to generate a set of intermediate key–value pairs.
2. In the reduce phase, the intermediate key–value pairs are grouped by key and the values are combined together according to the reduce algorithm provided by the user. Sometimes no reduce phase is required, given the type of operation coded by the user. MapReduce processes are divided between two applications, JobTracker and TaskTracker at the cluster level. JobTracker is responsible for scheduling job runs and managing computational resources across the cluster; hence it runs on only one node of the cluster. Each MapReduce job is split into a number of tasks which are assigned to the various TaskTrackers depending on which data is stored on that node. So TaskTracker runs on every slave node in the cluster. JobTracker oversees the progress of each TaskTracker as they complete their individual tasks.



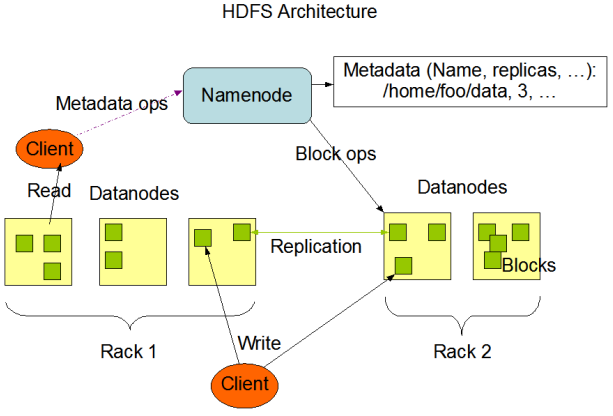
The main components of MapReduce are listed below:

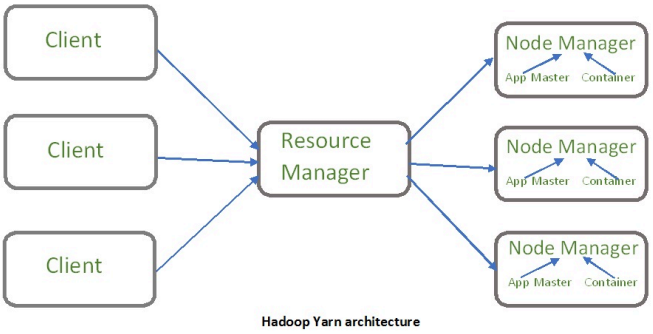
1. JobTrackers: JobTracker is the master which manages the jobs and resources in the cluster. The JobTracker tries to schedule each map on the TaskTracker which is running on the same DataNode as the underlying block.

2. TaskTrackers: TaskTrackers are slaves which are deployed on each machine in the cluster. They are responsible for running the map and reduce tasks as instructed by the JobTracker.

3. JobHistoryServer: JobHistoryServer is a daemon that saves historical information about completed tasks/applications.

Yet Another Resource Negotiator (YARN) YARN addresses problems with MapReduce 1.0s architecture, specifically the one faced by JobTracker service. It splits up the two major functionalities “resource management” and “job scheduling and monitoring” of the JobTracker into two separate daemons. One acts as a “global Resource Manager (RM)” and the other as a “ApplicationMaster (AM)” per application. Thus, instead of having a single node to handle both scheduling and resource management for the entire cluster, YARN distributes this responsibility across the clus‘--





Hadoop Ecosystem

Apart from HDFS and MapReduce, the other components of Hadoop ecosystem are shown in Fig. The main ecosystems components of Hadoop architecture are as follows:

HBase "is an open-source, distributed, versioned, column-oriented store" that sits on top of HDFS. HBase is based on Google's Bigtable. HBase is based on columns rather than rows. This essentially increases the speed of execution of operations if they are need to be performed on similar values across massive datasets; for example, read/write operations that involve all rows but only a small subset of all columns. HBase does not provide its own query or scripting language, but is accessible through Java, Thrift and REST APIs.

Hive provides a warehouse structure for other Hadoop input sources and SQL-like access for data in HDFS. Hive's query language, HiveQL, compiles to MapReduce and also allows user-defined functions (UDFs). Hive's data model is based primarily on three related data structures: tables, partitions and buckets. Tables correspond to HDFS directories that are divided into partitions, which in turn can be divided into buckets.
HCatalog is a metadata and table storage management service for HDFS. HCatalog's goal is to simplify the user's interaction with HDFS data and enable data sharing between tools and execution platforms.

Pig is a run-time environment that allows users to execute MapReduce on a Hadoop cluster. Pig Latin is a high-level scripting language on Pig platform. Like HiveQL in Hive, Pig Latin is a higher-level language that compiles to MapReduce. Pig is more flexible with respect to possible data format than Hive due to its data model. Pig's data model is similar to the relational data model, but here tuples can be nested. For example, a table of tuples can have a table in the third field of each tuple. In Pig, tables are called bags. Pig also has a "map" data type, which is useful in representing semi-structured data such as JSON or XML."

Sqoop ("SQL-to-Hadoop") is a tool which transfers data in both ways between relational systems and HDFS or other Hadoop data stores such as Hive or HBase. Sqoop can be used to import data from external structured databases into HDFS or any other related systems such as Hive and HBase. On the other hand, Sqoop can also be used to extract data from Hadoop and export it to external structured databases such as relational databases and enterprise data warehouses.

Mahout is a scalable machine-learning and data-mining library. There are currently following four main groups of algorithms in Mahout:
1. Recommendations/Collective filtering.
2. Classification/Categorization.
3. Clustering.
4. Frequent item-set mining/Parallel frequent pattern mining.

ZooKeeper is a distributed service with master and slave nodes for storing and maintaining configuration information, naming, providing distributed synchronization and providing group services in memory on ZooKeeper servers. ZooKeeper allows distributed processes to coordinate with each other through a shared hierarchical name space of data registers called

znodes. HBase depends on ZooKeeper and runs a ZooKeeper instance by default.

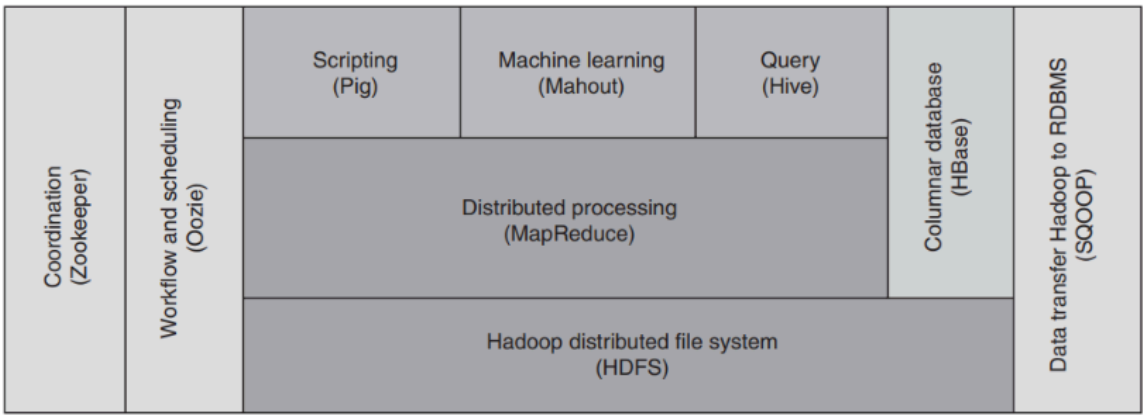


Figure 2.3 Hadoop ecosystem.

Hadoop Limitations

Security Concerns Hadoop security model is disabled by default due to sheer complexity. Whoever's managing the platform should know how to enable it; else data could be at huge risk. Hadoop does not provide encryption at the storage and network levels, which is a major reason for the government agencies and others not to prefer to keep their data in Hadoop framework.
Vulnerable By Nature Hadoop framework is written almost entirely in Java, one of the most widely used programming languages by cyber-criminals. For this reason, several experts have suggested dumping it in favor of safer, more efficient alternatives.
Not Fit for Small Data While big data is not exclusively made for big businesses, not all big data platforms are suitable for handling small files. Due to its high capacity design, the HDFS lacks the ability to efficiently support the random reading of small files. As a result, it is not recommended for organizations with small quantities of data.
Potential Stability Issues Hadoop is an open-source platform necessarily created by the contributions of many developers who continue to work on the project. While improvements are constantly being made, like all open-source software, Hadoop has stability issues. To avoid these issues, organizations are strongly recommended to make sure they are running the latest stable version or run it under a third-party vendor equipped to handle such problems.
General Limitations Google mentions in its article that Hadoop may not be the only answer for big data. Google has its own Cloud Dataflow as a possible solution. The main point the article stresses is that companies could be missing out on many other benefits by using Hadoop alone.

Feature	RDBMS (e.g., MySQL, Oracle)	Hadoop (Apache Hadoop)
Primary Purpose	Online Transaction Processing (OLTP)	Big Data Analytics & OLAP
Data Types	Strictly **Structured** (rows and columns)	**Structured, Semi-structured, & Unstructured** (videos, logs, XML)
Data Volume	Gigabytes to Terabytes	Terabytes to Petabytes and beyond
Scalability	**Vertical** (scale up by adding more CPU/RAM to one machine)	**Horizontal** (scale out by adding more servers to a cluster)
Schema	**Schema-on-Write** (rules enforced before data is saved)	**Schema-on-Read** (raw data is saved; schema applied when queried)
Data Integrity	High **ACID compliance** (guarantees transaction accuracy)	Base eventual consistency (optimized for throughput, not instant transactions)
Query Language	Standard SQL	HQL (Hive Query Language), MapReduce, or Spark
Hardware	Requires high-end, expensive servers	Runs on low-cost, commodity hardware